Chapter 11 | Written solutions

Long Context, Sparse Access, and Sequence Compression

Goal. Separate what a model stores, what it selects, and how attention weights the selected values. These twelve solutions progress from hand calculations to executable causal checks and a controlled experiment design. Selection can find the right row and still change the output; a summary can preserve a useful average while losing an exact fact.

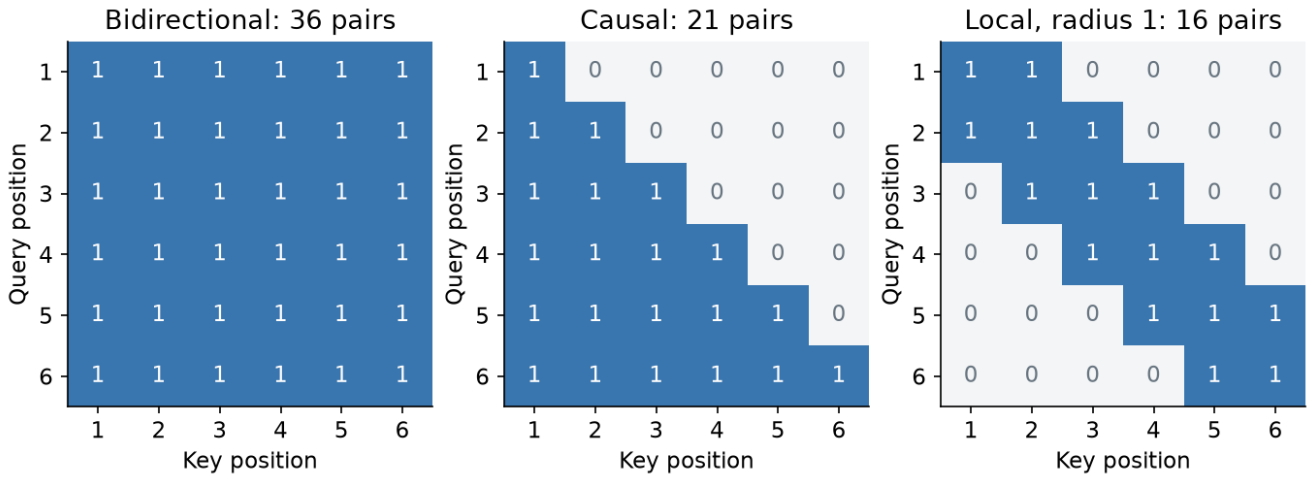
Setup. Open this notebook inside `transformers-book`, install `requirements.txt`, and run the cells in order with a Python kernel. All calculations run on CPU. Positions in Exercises 1–4 start at one; Python indices in Exercises 5–8 and 10 start at zero. The small NumPy models illustrate mechanisms rather than reproduce a production model.

Steps and checks. Work out the answer first, then use the recorded calculations and assertions to check it. Both companion PDFs are exported from this notebook: the written version contains every explanation and figure; the Python version also contains code and outputs.

1. Count the grid

Each row asks a question and each column identifies a key it may read. The diagonal is allowed: a token can read itself. With six positions, the full bidirectional grid has six reads in each of six rows, or $6 \times 6 = 36$ pairs. The causal grid has row counts $1, 2, 3, 4, 5, 6$, totaling 21. The last causal query can read every position, 1 through 6.

At radius one, local queries read their immediate neighbors and themselves. The first and last positions have only one neighbor; the four interior positions have two. Thus $2 + 4(3) + 2 = 16$ pairs. These are **logical permissions**. Constructing a dense grid and masking it does not by itself save the dot products.



The numbered cells make the boundary cases visible: the causal triangle grows down the rows, while the local band loses a neighbor at each end.

2. Trace a relay

Position 5 is global, so its query can read position 2 despite the gap. In the next layer, position 9 can read the updated representation at global position 5. If $h_i^{(\ell)}$ is the representation at position i after layer ℓ, the route is

$$h_2^{(0)} \longrightarrow h_5^{(1)} \longrightarrow h_9^{(2)}.$$

All queries in one layer read that layer's input representations. Position 9 cannot read the newly updated $h_5^{(1)}$ within layer 1; it must wait for layer 2. With local reads alone, each layer advances information by at most one position. The route $2, 3, 4, 5, 6, 7, 8, 9$ needs seven layers. A route permits information to flow; learned weights determine whether it is retained.

Both relay links also obey causality: query 5 reads earlier key 2, and query 9 reads earlier key 5. If only position 1 is global, a bidirectional encoder could use $2 \to 1 \to 9$, but a causal decoder cannot: query 1 is forbidden to read future key 2. The location of a global marker matters.

3. Choose, then weight

The index scores rank positions as $4, 1, 3, 2$. The top two are therefore position 4 (score 1.9) and position 1 (score 1.4), in that order. Their main scores are $(0, \log 2)$, so the exponentials are $(1, 2)$ and the weights are $(1/3, 2/3)$. Multiply these by the corresponding value rows:

$$\frac{1}{3}(3, 0) + \frac{2}{3}(0, 3) = (1, 2).$$

Position 4 ranked first during selection but receives the smaller main attention weight. The indexer decides which rows enter the calculation; the main scores determine their contributions.

4. Pool one coordinate

Multiply each feature by its weight and add the products:

$$0.25(4) + 0.5(0) + 0.25(2) = 1 + 0 + 0.5 = 1.5.$$

The different triple $(1, 2, 1)$ also gives $0.25 + 1 + 0.25 = 1.5$. A query receiving only this summary coordinate cannot distinguish the two source triples. Even if it knows the weights, it only knows the constraint $x_1 + 2x_2 + x_3 = 6$, which many triples satisfy. It cannot recover an exact position's feature or even the block's maximum: the examples have maxima 4 and 2.

This is the tradeoff of a summary. It may retain what a task needs, but the pooled number does not preserve every possible question about its sources.

5. Inspect a global mask

Imagine seven tokens at positions 0 through 6. A query at position 0 can use its own key and the key immediately to its right under a radius-one rule. Making position 3 global gives the query another place to look even though that key is three positions away. A **mask** records these permissions. In the Boolean array, row q belongs to query position q and column j belongs to key position j. A true entry lets that query read that key.

In the bidirectional rule, a pair is permitted when the query and key are at most one position apart or either position is global. This opens both row 3 and column 3. Causal mode first builds those same permissions and then keeps only the pairs with key position $j \le q$. The causal rule applies even to the global query and the global key.

For query 0, causality removes keys 1 and 3. Key 1 comes from the local window and key 3 comes from the global column, but both are in the future. For the global query at 3, causality removes keys 4, 5, and 6. The query can still read keys 0 through 3, including its own key. Query 6 keeps keys 3, 5, and 6 because all three are at or before it. Earlier keys 0, 1, 2, and 4 remain outside this query's radius and are not global. Causality removes permissions from the bidirectional grid and never adds a read.

6. Filter before gathering

The largest index score, 9.0, belongs to row 3. The allowed vector marks that row as false. The function first forms the candidate set $\{0, 1, 2\}$, then ranks those candidates by index score. The ranking is $[1, 0, 2]$ because $1.1 > 0.8 > 0.6$. Thus the first two picks are rows 1 and 0. When $k = 5$, only three candidates exist. The function uses all three and adds only row 2.

After gathering those keys and values, main attention computes a fresh score for each selected key. The score is the query–key dot product divided by the square root of the key width. Here the query is $[1]$, the width is one, and the score for each row is just its key coordinate. The index scores determine what to gather. The main scores determine how to mix the gathered values.

For $k = 2$, the selected main scores are $(1, 0)$ and their values are $(20, 10)$ in that order. Softmax exponentiates each score and divides by the sum of the exponentials. The weights and output are

$$w = \frac{(e, 1)}{e + 1} \approx (0.731059, 0.268941), \qquad y = \frac{20e + 10}{e + 1} \approx 17.310586.$$

For $k = 5$, the selected main scores are $(1, 0, 2)$, and the values are $(20, 10, 30)$. The newly admitted key gets score 2, the highest **main** score, even though it ranked last among the allowed candidates. The normalization now includes e^2, so the first two weights change as well.

$$w = \frac{(e, 1, e^2)}{e + 1 + e^2} \approx (0.244728, 0.090031, 0.665241), \qquad y = \frac{20e + 10 + 30e^2}{e + 1 + e^2} \approx 25.752104.$$

The output is a one-coordinate vector because each value row has width one. All positions below use zero-based indices. The weights are reported in **selected order**.

With $k = 5$, the gathered attention sees every allowed row. Its output therefore agrees with dense attention with the same mask. The dense weights are indexed in original row order, so gather those weights at the selected indices before comparing. The dense weight for row 3 is zero.

7. Compare summary windows

Imagine writing one short note for every two tokens. A stride of two determines when a new note is ready. An overlapping note can refer to the preceding block as well as the current one. Its end position still marks the same newest token. In this example the feature values are the integers 1 through 6, and their zero-based positions are 0 through 5.

A gate logit is a score that softmax turns into a weight. All gate logits are zero, so in a window of m positions each exponentiated score is $e^0 = 1$ and each weight is $1/m$. With one feature per token, each summary is just the mean of the values in its window.

End position	Positions without overlap	Mean	Positions with overlap	Mean
1	0, 1	$(1+2)/2 = 1.5$	0, 1	$(1+2)/2 = 1.5$
3	2, 3	$(3+4)/2 = 3.5$	0, 1, 2, 3	$(1+2+3+4)/4 = 2.5$
5	4, 5	$(5+6)/2 = 5.5$	2, 3, 4, 5	$(3+4+5+6)/4 = 4.5$

The first window ends at position 1 in either mode. It contains two tokens because there is no preceding block at the start. The next two overlapping windows contain four tokens each. The first window's weights are therefore $1/2$ per position, while each later overlapping window uses $1/4$ per position.

The output has three rows and one feature per row in both modes. The positions $[1, 3, 5]$ record the end of each window. Every completed stride block produces one summary, so the overlapping windows change the values and the source positions without increasing the number of summaries. An unfinished final stride block would produce no row.

8. Give coordinates different gates

Now each token has two features. The first coordinate reads $(2, 0, 6)$ across the three positions. The second reads $(0, 4, 2)$. With stride three there is one completed window, ending at position 2. Each coordinate forms its own weighted sum over that same window.

Softmax exponentiates the logit in each row and divides by the sum of exponentials in that coordinate's column. The first gate column contains the logarithms of $(1, 2, 1)$, so exponentiating recovers those numbers. Their sum is 4, and the weights are $(1/4, 2/4, 1/4)$. The first summary coordinate is

$$\frac{1}{4}(2) + \frac{2}{4}(0) + \frac{1}{4}(6) = 2.$$

The second gate column contains the logarithms of $(3, 1, 2)$. Its exponential sum is 6, giving weights $(3/6, 1/6, 2/6)$. The second coordinate is

$$\frac{3}{6}(0) + \frac{1}{6}(4) + \frac{2}{6}(2) = \frac{8}{6} = \frac{4}{3}.$$

The zero feature at position 1 receives half of the weight in the first coordinate. The zero at position 0 receives half in the second coordinate. Each contributes zero to its sum while still using some of the weight. The weights are determined by the gate logits, so a feature value of zero does not remove its position from the softmax.

The helper transposes the gate array before softmax because its softmax normalizes the last axis. That puts positions on the axis being normalized. It transposes the weights back to match the original feature array, multiplies entries coordinate by coordinate, and sums over positions. The comparison uses float64 and an absolute tolerance of 10^{-12} with zero relative tolerance.

9. Account for three designs

A stored main-attention KV entry consists of one key and its associated value. An index score ranks a candidate, and a main read supplies an entry to attention. For one query with 128 visible positions:

Design	Stored KV entries	Index scores	Main reads
Dense	128	0	128
Select eight raw entries	128	128	8
Read stride-eight summaries + four recent entries	20	0	20

Dense attention keeps and reads all raw positions. Selection still keeps all 128 because future queries may choose different rows; the indexer scans all of them before gathering eight. With stride eight there are $128/8 = 16$ completed summaries. Adding four raw entries gives $16 + 4 = 20$ stored entries and reads.

In one-based positions, the final summary covers 121–128 and the recent window covers 125–128. These overlapping sources do not make the cache entries identical: one summary and four raw entries occupy five separate rows. Count them separately.

These are main-cache row counts, not total system bytes. Actual storage depends on widths and precision. Persistent index features, partial-summary state, position metadata, and scratch buffers need their own accounting. Old raw data may be evicted only when all state needed for future summaries has been retained.

10. Check visibility and joint normalization

At query position 7, the decoder has seen the values 1 through 8. The third stride block starts at position 6, so the decoder can already see its first two values, 7 and 8. Its last value is at position 8, which comes after the query. To decide whether a query can read an entire summary, check the newest source position in that summary.

We can build all three summaries on the completed array and then ask which would be available to the earlier query. A zero gate array gives equal weights over each three-token window. Changing the value at position 8 from 9 to 90 changes only the third window. Its mean increases by $(90 - 9)/3 = 27$.

Source positions	End position	Original mean	New mean	Visible at query 7?
0, 1, 2	2	$(1 + 2 + 3)/3 = 2$	2	Yes
3, 4, 5	5	$(4 + 5 + 6)/3 = 5$	5	Yes
6, 7, 8	8	$(7 + 8 + 9)/3 = 8$	$(7 + 8 + 90)/3 = 35$	No

Causal attention permits a read when the summary's end is at or before the query. Comparing the end positions to 7 gives the same mask after the edit because the source positions of each window are unchanged.

For these nonoverlapping windows of width three, the start position is the end minus two. If we test starts instead, every start passes because 0, 3, and 6 are all at or before position 7. That admits the third summary even though it includes the unseen value at position 8.

The correct mask leaves the visible summary values at $(2, 5)$ in both runs. With the start-based mask the query could see a summary that moves from 8 to 35 in response to a future edit. Checking the end prevents that information leak. At position 8 the third block is complete and its summary can first be read. If a decoder also has a recent raw window, it can read the already-seen positions in the unfinished block through that window while the summary remains unavailable.

For the joint-normalization extension, the first summary has value 2. Raw positions 6 and 7 have values 7 and 8. All three main logits are zero, so one softmax assigns weight $1/3$ to each entry:

$$y = \frac{2+7+8}{3} = \frac{17}{3} \approx 5.666667.$$

Normalizing the summary branch separately gives 2. Normalizing the recent branch gives $(7+8)/2 = 7.5$. Averaging these outputs gives $(2+7.5)/2 = 4.75$. It assigns half the total weight to one summary and divides the other half between two raw entries. That is a different operation. The end-position test assumes causally computed features and gates; it cannot repair future information already mixed into them.

11. Measure missed evidence (optional)

Imagine one query looking for a fact that appears only in row 2. Give that row a positive first value coordinate, and set the first coordinate to zero on all four distractors. This coordinate lets us see the effect of omitting the target. Give distractors different second coordinates so changes to their relative mixture are also visible.

Use the fixed width-one query $[1]$ and keys of width one. Use natural logarithms of $(1, 2, 12, 3, 1)$ as the key coordinates. The main scores equal those coordinates. Full attention gives weights $(1, 2, 12, 3, 1)/19$ and output $(120/19, 18/19)$ with the values below. It assigns the largest weight to the target. We keep the query, keys, values, and available positions fixed for the entire sweep.

The separate indexer gives distractors at positions 0, 1, 3, and 4 fixed scores of 0.9, 0.7, 0.4, and 0.2. Sweep the target index score from 0.1 to 1.0 and try $k = 1, 2, 3, 5$. With a score of 0.4, row 2 ties row 3 and wins because its index comes earlier. The target is then ranked third and fits at $k = 3$. With a score of 0.7, it ties row 1 and loses that tie. A budget of two still misses the target. The main scores never change.

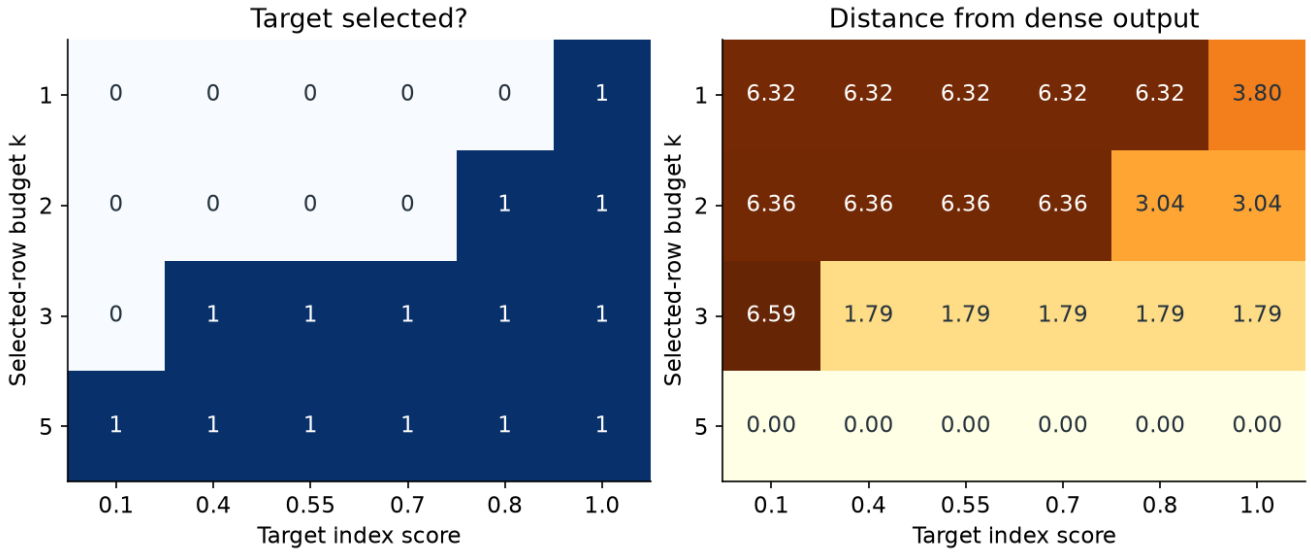
For every pair of target score and k, record the gathered positions and whether row 2 was among them. Compare the two-coordinate outputs using their Euclidean distance

$$\left\| y_{\text{selected}} - y_{\text{dense}} \right\|_2 = \sqrt{(y_1 - y_{\text{dense},1})^2 + (y_2 - y_{\text{dense},2})^2}.$$

This is a distance in output space. We also print the first coordinate so we can distinguish loss of the target from a change in weighting when the target stays selected.

A miss erases the target signal in the first coordinate. For example, with a target score of 0.1 and $k = 3$, that coordinate is zero while dense attention gives $120/19 \approx 6.316$. When the target score rises to 1.0 at $k = 1$, the target is the only retained value. Its first coordinate is now 10. This run finds the target but still differs from dense attention, which gives some weight to the distractors. Keeping the target at $k = 3$ with score 1.0 retains two distractors too. The first coordinate becomes $120/(12 + 1 + 2) = 8$. At score 0.1, raising k from one to three admits only distractors. The first coordinate stays zero and the second moves farther from the dense reference, so the distance grows in this case.

Two distinct mechanisms contribute to the output error. An omitted row contributes nothing. Even when the target is retained, softmax renormalizes across the rows that remain. With $k = 5$, all five rows contribute to both sums in softmax attention. Changing their order leaves the mathematical output unchanged and the NumPy results agree within the tolerance. This controlled fixture measures how ranking changes one attention output. A trained model's answer quality requires a separate test on labeled tasks.



The left panel records selection as 1 (yes) or 0 (no); the right panel shows Euclidean distance in output-vector units. A hit does not imply zero error. Only selecting all five rows guarantees agreement here, because it preserves the full softmax denominator and every value contribution. More selected rows need not monotonically reduce the distance for this fixed index ranking.

12. Plan a long-document comparison

One workable experiment uses a registry of shipments, archives, and seal codes. The one-fact task asks for the seal of shipment R17, whose record says it uses C42. The two-fact task separates the evidence: one record links R17 to archive A03; another links A03 to C42. Ask for both the archive and the seal in the second task so finding only one fact earns partial credit rather than a correct joint answer.

Construct documents of 2,048, 8,192, and 32,768 tokens with a common tokenizer and the question at the end. For length T, place the one-fact record at $T/8$. For two facts, use $T/8$ and $3T/8$ and counterbalance which record comes first. Create nearby controls at $T - 96$ and $T - 64$, using only $T - 64$ for one fact. Use short fixed-length templates, reserve room for the question, and record actual tokenized start and end positions so evidence cannot overlap the question or other evidence.

Fill the remaining document with plausible distractors. Randomize IDs and codes so the answer cannot be guessed from the question, position, or filler, and ensure each requested answer is unique. Pair near and far versions by moving the same records while preserving the question and answer. Split by underlying document **before** making variants; otherwise a near training example could leak into a far test example. Reserve 400 test documents per task and length, each with both placements. Every system receives the same test inputs and output instructions.

Use causal decoders with eight layers, equal width, the same tokenizer, positional convention, and output head. The context capacity must allow at least 32,800 tokens, including the continuation. Start shared parameters from paired initializations and train on the same documents and token budget. Fix architecture-specific settings using validation data before testing. A concrete set of policies is:

Method	History available to a query
Dense	All current and earlier raw positions
Local	Current position and up to 128 earlier positions per layer; no global markers
Learned selection	Up to 128 raw entries, ranked by scanning all eligible raw positions
Sequence summaries	All completed stride-16 summaries plus 128 recent raw entries

Train the indexer using an auxiliary ranking objective favoring annotated evidence for the answer query. Give all four models separate training prompts asking for the same supporting-record IDs, and disclose the selector's extra objective and training cost. Rank only eligible positions. For summaries, use causally computed features and require the summary's last position to be visible; count summary and raw entries separately when their source spans overlap. With eight local layers and radius 128, a path can advance at most $8(128) = 1,024$ positions, so nearby and distant evidence test different access conditions. These are separately trained architectures, not a claim that masks can be swapped without adaptation.

For answer quality, decode greedily until a newline or 32 generated tokens. Require `seal=CODE` for one fact and `archive=ID seal=CODE` for two facts. Normalize surrounding whitespace and case, then score exact match for the seal, each field separately, and both fields jointly. Missing or malformed fields are incorrect. Report results by task, length, and evidence placement, with paired differences on the same underlying documents. Bootstrap whole documents if estimating uncertainty so near/far variants stay paired. For selection, record whether a key from **each** required evidence record reaches the selected set at the answer query in a specified layer and head. This measures direct access at that point; answer accuracy measures behavior after all layers.

Measure memory and latency on the same GPU at batch size one. Use float16 inference where supported and disclose actual kernel precision, device, and software. After prefill and after decoding, count live persistent bytes from array shapes and dtype sizes across all layers: raw KV, summaries, partial-summary buffers, index keys, and position metadata. Report these components and allocated capacity separately. Also measure peak temporary and total memory, including index scans, dense score arrays, and gather buffers. Log index-score counts and main-read counts independently.

Time prefill from device-ready input IDs through logits for the first continuation token, including creation of summaries and index state. For decode timing, replay the **same** continuation of 32 token IDs for all methods. Prefill supplies the first prediction; feed the first 31 IDs and time those 31 incremental calls. Include cache updates, selection, gathering, and attention. This controlled timing workload is separate from the greedy quality test, whose generated tokens can differ. Warm up, synchronize the GPU at timing boundaries, rotate method order, and run 20 trials on the same prompts. Report median prefill latency and decode milliseconds per step. If a method runs out of memory, record that result; compare speed only for methods that complete the same workload.

No benchmark scores are invented here. This is an experimental protocol: it specifies what to hold fixed, what to measure, and which conclusions each measurement supports.

Checks and next steps. The executed assertions check causal masks, summary visibility, pooling, joint softmax, and dense agreement when every row is selected. Try moving the global marker or changing the target rank; predict the result before running it. Keep storage, index work, main reads, and output error separate. The next chapter studies data movement while preserving the attention operation.